

Report on Plans and Challenges faced

My report outlines the primary technical and analytical challenges encountered during the development of my bank deposit prediction model, detailing the specific techniques and strategic decisions I used to overcome them.

The first major hurdle was the technical environment setup, specifically the difficulty in installing and configuring the XGBoost library. This process was challenging due to compatibility requirements between the machine's architecture and the C++ compilers needed for the library's gradient boosting framework. To resolve this, I did some specialized installation paths and environment management. In the end this effort was justified because XGBoost ultimately outperformed the Random Forest baseline, providing me a more robust AUC of 0.8024 and superior precision, proving that the struggle to implement this specific tool was a critical investment for the project's success.

Data cleaning presented a significant decision regarding the "duration" variable. Initial analysis showed me that "duration" (the length of the call) was the strongest predictor of a "yes". However, I removed it from the model because its value is only known after a call is completed. Including it would have created data leakage, resulting in a model that looks perfect in testing but is useless for real-world prediction, as the bank cannot know how long a future call will last before making it. Furthermore, the dataset was surprisingly clean with almost no outliers, except for the campaign variable. I calculated and applied a percentile filter to campaign to remove extreme cases, ensuring the model didn't learn from rare "harassment" patterns that don't reflect standard banking policy.

A major analytical challenge was the shift in perceived feature importance. During the initial Exploratory Data Analysis (EDA), visual cues and raw counts suggested me that housing and loan (whether a customer had existing debts) would be the primary drivers of subscription. However, the machine learning models proved this intuition wrong. The final Feature Importance analysis revealed that macroeconomic indicators, particularly nr_employed (with a weight of ± 0.48) and euribor3m, were the true predictors. This shift highlighted the importance of using advanced algorithms to uncover deep, non-obvious correlations that human observation alone cannot detect in complex financial datasets.

To address the inherent class imbalance where "no" far outweighs "yes", I used SMOTE, but also a strategic threshold optimization technique plan. Rather than sticking to the default 0.5 probability, I tested a range of thresholds. The challenge was finding the "sweet spot" where we don't ignore too many potential customers. By analyzing the Precision-Recall trade-off, I found that a 0.3 threshold provides the best balance for a standard campaign. But it can be changed only changing the threshold value. This technical flexibility allows the model to be a dynamic business tool rather than a static mathematical formula.

And with my project I present a dynamic marketing strategy based on these threshold shifts. The model is designed to adapt to the bank's current objectives: if the marketing plan is aggressive (focused on high volume and market share), the threshold can be lowered to 0.2, pushing Recall to 0.58 to capture nearly 60% of all potential deposits. Conversely, if the plan is to be conservative (focused on cost-cutting and worker efficiency), the threshold can be raised to 0.8, ensuring a high Precision of 0.94. This adaptability ensures that the model remains a valuable asset regardless of shifting economic climates or changing corporate goals.

Thank you.